# Curriculum Vitae of Scientific and Teaching Activities

**Paolo Sorino**

July 29, 2026

## CV Summary

Paolo Sorino obtained his PhD in Information Engineering from Politecnico di Bari in 2025, defending a thesis titled "Leveraging Artificial Intelligence for Enhanced and Human-centered Healthcare Solutions" under the supervision of Tommaso Di Noia, with co-supervisors Fedelucio Narducci and Rodolfo Sardone.

He previously worked as a Research Grant Holder at IRCCS Saverio De Bellis in Castellana Grotte, where his research focused on artificial intelligence applications in epidemiology and biostatistics, particularly machine learning for healthcare.

Since March 2022, he has held consecutive Research Fellow positions (SSD ING-INF/05) at the Department of Electrical and Information Engineering (DEI) of Politecnico di Bari, contributing to national and European projects including PIA (chatbots and AI solutions), MISTRAL (health impact analysis and disability-related costs prediction), LIFE (Italian system-wide Frailty network), OMIBREED, CALLIOPE, and REACH-XY, applying AI, big data, and XAI techniques to healthcare and agritech challenges. He serves as Work-Package Leader in multiple national research projects. In 2026, he was awarded an ISCRA-C project as Principal Investigator through the Italian SuperComputing Resource Allocation programme managed by CINECA, obtaining competitive access to national High-Performance Computing (HPC) resources for research in Artificial Intelligence and Data Science.

His research interests span ML/DL, biomedical informatics, healthcare AI, GNN, knowledge graph embeddings, XAI, ECG analysis, cardiac signal processing, and BCI systems. His work addresses clinical prediction, drug-disease interactions, and explainable models in medicine, resulting in 41 peer-reviewed publications (20 journal articles + 21 conference/workshop papers).

He holds an Italian national patent on a ML-based method for validating NAFLD diagnosis without imaging technologies and serves as Guest Editor for Sensors (MDPI) Special Issue on "Advances in Sensorized AI-Driven Intelligent Systems in Healthcare and Beyond".

Since 2022, he has been intensively involved in research service as PC member, workshop chair, and organizer at major conferences (IEEE SMC, ACM UMAP, ICWE, ICHMS, RecSys, ECIR) and as reviewer for top-tier journals (Neurocomputing, IEEE THMS, Scientific Reports, BMC Medical Informatics, Expert Systems with Applications).

Since 2022, he has provided continuous teaching support (SSD ING-INF/05) at Politecnico di Bari across Computer Engineering programs. He served as Assistant Lecturer (2022–2025: Foundations of Machine Learning; 2023–2024: Deep Learning; 2024: Computer Architectures), Subject Expert (2025), and Adjunct Professor (2025–2026: Software Design Laboratory; 2026: 2nd-Level Master's Degree in Artificial Intelligence and Data Science). He has delivered more than 200 hours of specialized training for PNRR programs, high schools, ITS Apulia Digital, and professional institutions.

# CURRICULUM INDEX

# 1 GENERAL INFORMATION

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## 1.1 Personal Data

### 1.1.1 Current Affiliation

**Politecnico di Bari** Department of Electrical and Information Engineering (DEI)
Via Edoardo Orabona, 4, 70125 Bari (Italy)
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Web: https://sisinflab.poliba.it/people/paolo-sorino/
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## 1.2 Current Position

Postdoctoral Researcher in Information Processing Systems (SSD ING-INF/05) at Politecnico di Bari.

## 1.3 Previous Positions

- *March 2024 – Present,*

  **Postdoctoral Researcher** scientific-disciplinary sector ING-INF/05, at the Department of Electrical and Information Engineering (DEI), Politecnico di Bari.
  Project: LIFE - "the itaLian system-wIde Frailty nEtwork" (Italian Ministry of Health - Trajectory 2 "E-Health, advanced diagnostics, medical devices and mini-invasiveness").
  State-of-the-art analysis and development of AI and big data solutions for healthcare applications.

- *March 2023 - March 2024,*
  **Research Fellow** scientific-disciplinary sector ING-INF/05, at the Department of Electrical and Information Engineering (DEI), Politecnico di Bari. Project: MISTRAL - European Horizon 2020 project.

- *March 2022 - March 2023,*
  **Research Fellow** scientific-disciplinary sector ING-INF/05, at the Department of Electrical and Information Engineering (DEI), Politecnico di Bari. Project: PIA - Artificial Intelligence (AI), big data and chatbot solutions.

- *March 2019 - February 2022,*
  **Research Grant Holder** at IRCCS "Saverio De Bellis" Castellana Grotte Research Hospital. Research activities focused on AI, particularly in the field of Machine Learning (ML) applied to epidemiology and biostatistics.

- *January 2019 - Present,*
  **Technical and Scientific Collaborator** at Prime Engineering (Eng. Giovanni Pascoschi).

### 1.4 Academic Qualifications and Certifications

- *2025*
  **PhD in Information Engineering** (XXXVII cycle) obtained from the Department of Electrical and Information Engineering (DEI), Politecnico di Bari. Level 8 EQF.
  Thesis title: *"Leveraging Artificial Intelligence for Enhanced and Human-centered Healthcare Solutions"*.
  Supervisor: Prof. Tommaso Di Noia; Co-supervisors: Prof. Fedelucio Narducci, Dr. Rodolfo Sardone.

- *2021*
  **Professional Engineering License**, obtained at Politecnico di Bari (II session 2021).

- *2021*
  **Master's Degree in Computer Engineering** (LM-32 D.M. 270/04), obtained from Università degli Studi Guglielmo Marconi, Rome. Level 7 EQF, Grade: 108/110.
  Thesis title: *"An explainable artificial intelligence approach applied to a neural network to identify patients at risk of non-alcoholic fatty liver disease (NAFLD)"*.

- *2019*
  **Bachelor's Degree in Computer and Automation Engineering** (L-8 D.M. 270/04), obtained from Politecnico di Bari. Level 6 EQF, Grade: 80/110.
  Bachelor's thesis in Software Engineering: *"Use of Machine Learning techniques as support for the diagnosis of non-alcoholic fatty liver disease (NAFLD)"*.

- *2014*
  **High School Diploma in Computer Science** (Grade: 100/100), obtained from I.I.S.S. "Vito Sante Longo", Monopoli.

- *2021*
  **Class A Amateur Radio Operator License**, issued by the Italian Ministry of Economic Development.

- *2019*
  **Course "Machine Learning in STATA"**, participation in the training course "Machine learning in Stata: an introduction".

- *2014*
  **ECDL**, European Computer Driving Licence certification.

- *2012*
  **Trinity College Certificate**, Trinity College London certificate, Level A2.2 in English language.

# 2 SCIENTIFIC ACTIVITY

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Research activity, initiated in 2021 immediately after obtaining the Master's degree, has been developed mainly at Politecnico di Bari within the research group of the Information Systems Laboratory (SisInfLab `http://sisinflab.poliba.it`), and in collaboration with IRCCS "Saverio De Bellis" Research Hospital.

Within SisInfLab, research activity has focused on AI, ML, Natural Language Processing (NLP), Information Security and Data Science applied to complex information systems. The work has been carried out in a highly interdisciplinary and collaborative context, contributing to both national and international research projects. In particular, the activity has concerned the design and implementation of AI-based solutions for semantic analysis, knowledge extraction, and data modeling in digital ecosystems. Research conducted within the laboratory has combined theoretical investigation and experimental validation, including dataset construction, model fine-tuning, performance evaluation, and dissemination of results through scientific publications and conference contributions.

The overall research path has been characterized by a continuous integration between scientific rigor and applicative impact, combining methodological research and applied innovation in the fields of AI and intelligent information systems, with particular emphasis on healthcare applications.

## 2.1 Research Lines

Research lines and related topics can be grouped as follows:

- **ML and Deep Learning (DL) for Healthcare**: Application of advanced ML and DL techniques to clinical prediction, risk assessment, disease classification, and patient stratification. Development of predictive models for conditions such as NAFLD, MAFLD, Alzheimer's disease, cognitive impairment, metabolic syndrome, and cardiovascular diseases. Integration of multi-modal data (clinical, imaging, omics, biomechanical) for comprehensive patient profiling.

- **Explainable Artificial Intelligence (XAI) for Clinical Decision Support**: Development of interpretable and explainable AI models to enhance transparency and trust in clinical decision-making. Application of techniques such as SHAP, LIME, attention mechanisms, and saliency maps to elucidate model predictions. Focus on regulatory compliance and ethical considerations in medical AI.

- **Brain-Computer Interface (BCI) and Neuroscience Applications**: Design and implementation of BCI systems for emotion recognition, mental state classification, motor imagery, and neurorehabilitation. EEG signal processing and analysis using advanced ML and DL architectures. Development of novel datasets and benchmarking frameworks for BCI research.

- **Graph Neural Network (GNN) and Knowledge Graph Embeddings**: Application of GNN architectures (GCN, GAT, GIN) and knowledge graph embedding techniques (Dist-

Mult, ConvE, TransE) to biomedical knowledge representation, drug-disease interactions, and clinical pathway modeling. Integration of structured and unstructured biomedical data.

- **ECG Analysis and Cardiac Signal Processing**: Development of automated ECG classification systems for arrhythmia detection and cardiac abnormality identification. Application of time-series analysis, signal processing techniques, and deep learning architectures (CNN, RNN, LSTM, Transformers) to cardiac data.
- **Natural Language Processing in Healthcare**: Text mining and information extraction from clinical notes, electronic health records, and biomedical literature. Application of transformer-based models (BERT, GPT) for clinical entity recognition, relation extraction, and medical question answering.
- **Multi-modal AI Frameworks**: Integration of heterogeneous data sources (clinical, imaging, omics, wearable sensors) for comprehensive patient assessment. Development of fusion architectures and attention mechanisms for effective multi-modal learning.
- **Federated Learning and Privacy-Preserving AI**: Design of federated learning frameworks for healthcare applications that preserve patient privacy while enabling collaborative model training across institutions. Exploration of differential privacy and secure multi-party computation techniques.

The results obtained are documented in over 40 scientific publications, including international journal articles and conference and workshop papers.

## 2.2 Participation in Research Groups

Since 2019, research activities have been developed within collaborative frameworks involving national and international research groups:

- **SisInfLab - Information Systems Laboratory**, Politecnico di Bari: Core research group focusing on AI, ML, knowledge representation, and intelligent information systems.
- **Unit of Research Methodology and Data Sciences for Population Health**, IRCCS "Saverio de Bellis" Research Hospital, directed by Prof. Rodolfo Sardone (2019-2022): Collaboration on ML applications in healthcare, including (i) analysis of biomechanical data from IoMT devices; (ii) construction of dietary patterns associated with diabetes in elderly populations of Southern Italy; (iii) study of nutritional gaps for targeted management of aging populations.
- International collaborations with researchers from:
  - Hamad Bin Khalifa University, Doha (Qatar)
  - Eindhoven University of Technology (Netherlands)

  - Shandong University of Technology (China)
  - University of Suffolk, Ipswich (United Kingdom)
  - AGH University, Kraków (Poland)
  - University of Hasselt (Belgium)
  - University of Liverpool (United Kingdom)
  - CentraleSupélec, Inria, Université Paris-Saclay (France)

- Industrial research collaborations with companies such as QuestIT, and Servizi Locali as well as participation in Regional Technology Districts and Poles.

## 2.3 Participation in Research Projects

Research has been typically framed within international and national projects (both basic and industrial research), briefly described below.

### 2.3.1 National Projects

- *2026* **ISCRA-C Project – CINECA High Performance Computing (HPC). Principal Investigator** of a research project selected through the Italian SuperComputing Resource Allocation (ISCRA) programme. Granted access to national High-Performance Computing (HPC) infrastructure at CINECA for large-scale experimentation and computational research in Artificial Intelligence and Data Science. Duration: 9 months.

- *2024 - 2025*. Participation in project **OMIBREED** - "Characterisation and valorisation of agrobiodiversity through multi-omics and next generation breeding approaches for resistance to *Xylella fastidiosa*". Coordinated by CNR-IPSP; funded by MASAF (CUP B93C22001440005). Total budget: EUR 3,390,735.00; DEI unit funding: EUR 184,240.00.
  **Role**: Work-Package Leader WP3 "High-throughput phenotyping with integration of omics technologies for the study of resistance mechanisms to Xf and agronomic traits". Responsibilities include development of AI and XAI pipelines for genotype-phenotype-multi-omics correlation, biomarker identification, varietal classification models, and bioinformatics platform design.

- *2024 - Present*. Participation in project **CALLIOPE** (Casa delle Tecnologie Emergenti, Taranto) - "Casa dell'Innovazione per il One Health". Coordinated by Municipality of Taranto; funded by Italian Ministry of Enterprises and Made in Italy (CUP F89J21001230001). Total budget: EUR 13,871,930.21; DEI unit funding: EUR 183,750.00.
  **Role**: Work-Package Leader for WP1 "One Health Intelligence Department" (AI, Federated Learning, XAI) and WP2 "Innovation Technology & Deployment Hub" (intelligent robotics, IoT sensing, 5G/6G connectivity). Activities include AI/XAI model implementation, integrated web platform development, and technical-scientific coordination.

- *2023 - 2025.* Participation in project **REACH-XY** - "Research actions for reducing the impact on agricultural and natural ecosystems of the harmful plant pathogen *Xylella fastidiosa*". Coordinated by CNR-IPSP; funded pursuant to Law 234/2021 (CUP B93C22001920001). Duration: 4 years. Total budget: EUR 7,508,433.20; DEI unit funding: EUR 149,650.00.
  **Role**: Work-Package Leader for WP1 "Innovation-Oriented Collaborative Research Infrastructures in Plant Health", WP2 "Advanced Technological Approaches for Early Detection" (ML/DL/XAI for diagnosis), and WP4 "Innovative Biological Treatments and Rational Drug Discovery". Activities include AI/XAI model development, interactive web platform design, and technical-scientific coordination.

- *2023 - 2027.* Participation in project **POS - LIFE** - "the itaLian system wIde Frailty nEtwork". Italian Ministry of Health, Trajectory 2 "E-Health, advanced diagnostics, medical devices and minimally invasive technologies" (T2-AN-12, CUP POLIBA D93C22000640001). Contribution to the development and deployment of a Virtual Infrastructure for federated data access and processing in a Global As View perspective.

- *2023 - Present.* Participation in European project **MISTRAL** - "a toolkit for dynaMic health Impact analysiS to predicT disability-Related costs in the Aging population based on three case studies of steeL-industry exposed areas in europe". Horizon 2020, grant agreement no 101095119. Contribution to the design of virtual interoperability infrastructure architecture, particularly regarding the Security Module for cloud-based monitoring, threat detection, and endpoint protection.

- *2023 - 2025.* Participation in project **DEMETRA** - "Development of an ensemble learning-based, multi-dimensional sensory impairment score to predict cognitive impairment in an elderly cohort of Southern Italy". PNRR-MAD-2022-12376656, PNRR: M6/C2. Collaboration on software development, data management, process engineering, and AI-based model coordination.

## 2.4 Competitive Research Grants and Computing Resources

- 2026 – ISCRA-C Project, CINECA HPC. Principal Investigator. Competitive allocation of national High-Performance Computing resources through the Italian SuperComputing Resource Allocation programme.

## 2.5 Research Prototypes

Research activity has been consistently accompanied by significant prototyping development (software and hardware/software), conducted personally and supervising thesis students and collaborators, in order to validate the proposed models, techniques and algorithms. Below is a brief description of the most recent and significant prototypes that have been developed and are currently under maintenance and further development:

- **MORIX** [2025]: AI-based framework for mortality risk prediction in MAFLD patients. Uses epidemiological data, feature selection, ML model training, XAI analysis (SHAP), and web application for real-time predictions and explanations.
- **NeuroSense** [2024]: Novel EEG dataset utilizing low-cost, sparse electrode devices for emotion exploration. Includes data acquisition pipeline, preprocessing methods, and benchmark ML/DL models.
- **EEGScope** [2025]: Framework for real-time acquisition, processing, and analysis of EEG data for BCI applications. Supports multiple EEG devices, implements signal processing algorithms, and provides visualization tools.
- **NeuralPMG** [2024]: Neural Polyphonic Music Generation system based on ML algorithms for BCI-controlled music composition and neurorehabilitation.
- **ECG Classification System** [2024]: Explainable ML approach for heartbeat classification through signal-based features, including arrhythmia detection and cardiac abnormality identification.
- **Clinical Prediction Web Applications**: Series of web-based applications for clinical prediction tasks, including cognitive impairment assessment, diabetes risk stratification, and metabolic syndrome management.

## 2.6 Research Service Activities

### 2.6.1 Guest Editor

- 2025 - Guest Editor of the Special Issue "Advances in Sensorized AI-Driven Intelligent Systems in Healthcare and Beyond" in Sensors (MDPI) Journal (Q1, Impact Factor 3.5, CiteScore 8.2), Section Intelligent Sensors.

### 2.6.2 Conference and Workshop Organization

Active participation in the organization of scientific events, including conferences and workshops of national and international scope. List of organized events:

- **Invited Speaker**, 2026 IEEE SMCS International Summer School on Artificial Intelligence for Cognitive Technologies (ISACT 2026), part of the 2026 IEEE International Conference on Human-Machine Systems (ICHMS 2026), lecture "AI for BCI with applications", Nanyang Technological University, Singapore, July 1–3, 2026.
- **Mini-School Lecturer**, WOA 2026 — 27th Workshop "From Objects to Agents", Salerno, Italy, June 17, 2026. Lecture: Artificial Intelligence and Explainable AI in Clinical Decision Support Systems: Innovation, Interpretability, and Trust.
- **TPC Member** - IEEE ATC 2026 at IEEE Smart World Congress 2026.

- **PC Member and PhD Forum Chair** - IEEE SMC 2026 16th Workshop on Brain-Machine Interface (BMI) Systems.
- **Invited Speaker** at IEEE System Man and Cybernetics (SMC2025) 15th Workshop on Brain-Machine Interface (BMI) Systems (BMI 2025), titled "Brain Computer Interfaces for Neural Games" within the special event "Integrative Approaches to EEG Signal Analysis: Collaborative Perspectives from Multiple Disciplines".
- **PC Member**- CIKM2026, 35th International ACM Conference on Knowledge and Information Management.
- **PC Member** – RecSys 2026 (Reproducibility Track), 19th ACM Conference on Recommender Systems.
- **PC Member** - ECIR 2026 Reproducibility Track, 48th European Conference on Information Retrieval.
- **PC Member** - The Web Conference 2026 (WWW 2026), track "Search and Retrieval-Augmented AI".
- **PC Member, Workshop Chair and Workshop Organizer** - 2nd Workshop on Wearable Devices and BCI for User Modelling (WeBIUM) at the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP 2025).
- **PC Member** - RecSys 2025 (Reproducibility Track), 19th ACM Conference on Recommender Systems.
- **PC Member** - ECIR 2025 Reproducibility Track, 47th European Conference on Information Retrieval.
- **PC Member and Co-Organizer** - IEEE ICHMS 2025 Special Session on "Brain-Computer Interfaces, Large Language Models and Explainable Artificial Intelligence for computing in intelligent systems".
- **Local Organisation Chair** - Brain Informatics 2025 (BI 2025) - "Brain Science meets Artificial Intelligence", Bari, Italy.
- **Session Chair** - International Conference on Brain Informatics (BI 2025), track "Human Information Processing and Pain".
- **PC Member, Workshop Chair and Workshop Organizer** - 1st Workshop on Wearable Devices and BCI for User Modelling (WeBIUM) at the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP 2024), Cagliari, Sardinia, Italy.
- **PC Member** - International Workshop on Web Applications for Life Sciences (WALS 2024) at the International Conference on Web Engineering (ICWE 2024).

- **PC Member** - RecSys 2024 (Reproducibility Track and KaRS: Knowledge-aware and Conversational Recommender Systems Workshop).
- **PC Member** - IEEE ICHMS 2024 (Regular Papers), 4th IEEE International Conference on Human-Machine Systems.
- **PC Member** - RecSys 2024 KaRS Workshop: Sixth Knowledge-aware and Conversational Recommender Systems Workshop.
- **Participation and Paper Presentation** - International Conference on Web Engineering (ICWE 2022), Bari, Italy, July 5–8, 2022. Doctoral Symposium.
- **Invited Speaker** - Workshop "The potential of enabling technologies in tailoring and adapting neuromotor and cognitive rehabilitation in children" at IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2022), Ioannina, Greece, September 27–30, 2022 (Organizers: Dr. Ilaria Bortone, Dr. Lucia Billeci).
- **Session Chair and Paper Presentation** - IEEE International Conference on Systems, Man, and Cybernetics (SMC 2022), Prague, Czech Republic, October 9–12, 2022. Track: "Learning to Optimize in Intelligent Systems II".
- **Paper Presentation** - IEEE International Conference on Systems, Man, and Cybernetics (SMC 2023), Maui, Hawaii, October 1–4, 2023.
- **Paper Presentation** - The 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP 2024), Cagliari, Sardinia, Italy, July 1–4, 2024.
- **Invited Speaker** - International School on Artificial Intelligence for Cognitive Technologies (ISACT 2024), lecture "Artificial Intelligence for Brain-Computer Interaction", funded by the IEEE SMC Society, co-organised by Otto von Guericke University Magdeburg and the University of Naples Federico II, December 10–13, 2024.

### 2.6.3 Article Review Activities

Review activities for international workshops, conferences and journals. List of reviewing activities:

- Reviewer for Computers in Human Behavior Report (Q1)
- Reviewer for Neurocomputing(Q1)
- Reviewer for IEEE Transactions on Human-Machine Systems (Q1)
- Reviewer for Scientific Reports (Q1)
- Reviewer for Discover Artificial Intelligence (Q1)
- Reviewer for BMC Medical Informatics and Decision Making (Q1)

- Reviewer for npj Digital Medicine (Q1)
- Reviewer for Expert Systems with Applications (Q1)
- Reviewer for Journal of Ambient Intelligence and Humanized Computing (Q2)
- Reviewer for SMC 2022, SMC 2023, SMC 2024, SMC 2025
- Reviewer for RecSys 2022, RecSys 2024, RecSys 2025
- Reviewer for HT 2022 (ACM Conference on Hypertext and Social Media)
- Reviewer for CHI 2023 (ACM Conference on Human Factors in Computing Systems)
- Reviewer for ECAI 2024, ECAI 2025
- Reviewer for UMAP 2024, UMAP 2025
- Reviewer for ICWE 2024 (WALS Workshop)
- Reviewer for ECIR 2025, ECIR 2026 (Reproducibility Tracks)
- Reviewer for IEEE ICHMS 2025, IEEE ICHMS 2026
- Reviewer for IEEE CAI 2026

## 2.7 Patent

- **Italian National Patent** – Method for validating the diagnosis of non-alcoholic fatty liver disease (NAFLD) in patients under care without resorting to imaging technologies, using a machine learning algorithm.
  Patent number: 102020000030767. Publication number: IT 202000030767 A1.
  Inventors: Paolo Sorino, Alberto Rubén Osella. Assignee: IRCCS "Saverio de Bellis", Castellana Grotte (BA), Italy.
  International classification: G16H 50/20 – Health informatics.
  Filing date: 14 December 2020. Publication date: 14 June 2022. Grant date: 20 December 2022. Granting authority: Italian Ministry of Enterprises and Made in Italy (UIBM).
  The patent describes a non-invasive, low-cost decision support method for validating NAFLD diagnosis by means of a neural network trained on anthropometric and biochemical parameters, avoiding the use of imaging technologies.

- **Italian National Patent Application** – Neural control of self-propelled devices.

  Patent application no. 102024000005926. Inventors: Tommaso Colafiglio, Paolo Sorino, Domenico Lofù, Tommaso Di Noia. Filing date: 18 March 2024. Granting authority: Italian Ministry of Enterprises and Made in Italy (UIBM). Status: Accepted; grant pending.

The patent application describes a brain–computer interface (BCI)-based method and system for the neural control of self-propelled devices, enabling users to operate autonomous or semi-autonomous platforms through neural signals acquired from non-invasive brain activity.

## 2.8 Coordination, Supervision and Research Training

Continuous research training activities, coordinating and supervising bachelor's and master's students in experimental research thesis activities (from 2019 to present). Successfully supervised over 25 bachelor's and master's thesis projects in areas including:

- AI and ML applications in healthcare
- BCI and EEG signal processing
- XAI for clinical decision support
- Deep learning for medical imaging
- E-commerce platform development
- Database design and management

## 2.9 Dissemination of Research Activities

Since the beginning of the PhD path, regular international research presentation activities have been conducted.

### 2.9.1 Invited Talk

Since 2022, he has received the following **invitations** for **speaker** activities at International Workshops, organised at the conferences listed below:

1. 2026 - **Invited talk** at the IEEE SMCS International Summer School on Artificial Intelligence for Cognitive Technologies (ISACT 2026), organized within the IEEE International Conference on Human-Machine Systems (ICHMS 2026), Nanyang Technological University, Singapore, July 1–3, 2026.
   Title of the lecture: *"AI for BCI with Applications"*.
   [Link: `https://isact-org.github.io/`]

2. 2026 - **Mini-School Lecturer** at WOA 2026 – 27th Workshop "From Objects to Agents", Salerno, Italy, June 17, 2026.
   Lecture title: *"Artificial Intelligence and Explainable AI in Clinical Decision Support Systems: Innovation, Interpretability, and Trust"*.
   [Link: `https://sites.google.com/view/woa2026`]

3. 2025 - **Invited talk** at the IEEE System Man and Cybernetics (SMC) 15th Workshop on Brain-Machine Interface (BMI) Systems (BMI 2025), titled *"Brain Computer Interfaces for Neural Games"* on the *"Integrative Approaches to EEG Signal Analysis: Collaborative Perspectives from Multiple Disciplines"* special event.
   [Link: https://sites.google.com/view/smc-bmi-workshop2025/]

4. 2024 - **Invited talk** at the International School on Artificial Intelligence for Cognitive Technologies, (ISACT 2024), titled *"Artificial Intelligence for Brain-Computer Interaction"*. The School was funded by the IEEE SMC Society, organised by Otto von Guericke University (OvGU) Magdeburg, and co-organised and hosted by the University of Naples Federico II. December, 10-13, 2024.
   [Link of the School: https://isact-org.github.io/]

5. 2022 - **Invited Speech** at Workshop: '*The potential of enabling technologies in customising and adapting neuromotor and cognitive rehabilitation in children*'.
   Conference: *'International Conference on Biomedical and Health Informatics (BHI-BSN 2022)*'.
   Title of Speach: *'Composizione musicale in tempo reale basata sul riconoscimento delle emozioni AI'*.

### 2.9.2 Conference Paper Presentations

Regular speaker activities at international congresses since 2022. Presented papers include:

1. 2025 - International Conference on Brain Informatics (BI 2025), Session Chair for track "Human Information Processing and Pain".
   Session: "Human Information Processing and Pain".

2. 2024 - The 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP 2024), Cagliari, Italy.
   Paper presented: "ARIEL: Brain-Computer Interfaces meet Large Language Models for Emotional Support Conversation";
   Workshop organization: "WeBIUM 2024 - 1st Workshop on Wearable Devices and Brain-Computer Interfaces for User Modelling".

3. 2023 - IEEE International Conference on Systems, Man, and Cybernetics (SMC 2023), Maui, Hawaii.
   Papers presented: "A Pareto-Optimality-Based Approach for Selecting the Best Machine Learning Models in Mild Cognitive Impairment Prediction" and
   "Combining Mental States Recognition and Machine Learning for Neurorehabilitation".

4. 2022 - IEEE International Conference on Systems, Man, and Cybernetics (SMC 2022), Prague, Czech Republic. Session Chair and paper presentation.

Track: "Learning to Optimize in Intelligent Systems II";
Paper presented: "Brain Computer Interface: Deep Learning Approach to Predict Human Emotion Recognition".

5. 2022 - International Conference on Web Engineering (ICWE 2022) Doctoral Symposium, Bari, Italy.
   Doctoral research presentation: "Blockchain and AI to Build an Alzheimer's Risk Calculator".

# 3 TEACHING ACTIVITY

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Since 2021, continuous teaching support activities have been provided within the scientific-disciplinary sector ING-INF/05 for the Computer Engineering degree program at Politecnico di Bari.

## 3.1 Current Teaching Positions

- *2026*
  **Adjunct Professor** for the 2nd-Level Master's Degree in *Artificial Intelligence and Data Science*, Politecnico di Bari. Responsible for Module E, including lectures, laboratory sessions, and tutoring activities, for a total of 200 teaching hours.

- *2026*
  **Adjunct Professor** for the course "Software Design Laboratory (L-Z)" (6 CFU) (SSD ING-INF/05); Bachelor's degree in Medical Systems Engineering, Politecnico di Bari.

- *2025*
  **Adjunct Professor** for the course "Software Design Laboratory (L-Z)" (6 CFU) (SSD ING-INF/05); Bachelor's degree in Medical Systems Engineering, Politecnico di Bari.

## 3.2 Previous Teaching Activities

- *2025*
  **Subject Expert** for the following courses:
  - Foundations of Machine Learning (Prof. Tommaso Di Noia)
  - Human-Machine Interaction (Prof. Carmelo Ardito)
  - Computer Architectures (Prof. Domenico Lofù)

- *2022-2025*
  **Assistant Lecturer** for the course "Foundations of Machine Learning", Master's degree in Computer Engineering (Prof. Tommaso Di Noia), Politecnico di Bari.

- *2023-2024*
  **Assistant Lecturer** for the course "Deep Learning", Master's degree in Computer Engineering (Prof. Vito Walter Anelli), Politecnico di Bari.

- *2024*
  **Assistant Lecturer** for the course "Computer Architectures", Bachelor's degree in Computer and Automation Engineering - Taranto campus (Prof. Domenico Lofù), Politecnico di Bari.

- *2024*
  **Teaching Tutor for ANCI Courses** delivered by Politecnico di Bari (81 hours total):
  - Business Intelligence in Public Administration Process Management
  - Cybersecurity in Public Administration
  - Web Technologies for Public Administration
- *2024*
  **Training Lecturer** for the course "Automatic Learning for Data Analysis in Critical Contexts". Course delivered by DEI, Politecnico di Bari, within PNRR project "SCUOLA_UNI_PNRR" (15 hours). Delivered at Licei G. Galilei - M. Curie, Polo Liceale di Monopoli.
- *2024*
  **Training Lecturer** for the course "Fundamentals of Network and Systems Security; UF4 Security of Systems and Information" (10 hours). Delivered at Spegea Business School.
- *2024*
  **Occasional Consultant** (19/09/2024) - Lecturer for training course "AI applied to clinical data – When data become care: the path towards innovative diagnosis". Assinter Italia.
- *2024*
  **Occasional Consultant** (19/03/2024) - Awarded contract for supplementary, preparatory and remedial teaching activities for a.y. 2022-2023 for the course "Computer Architectures" (40 hours). Politecnico di Bari.
- *2025*
  **Training Lecturer** for the course on Artificial Intelligence (ATT-827), within STEM, digital and innovation skills strengthening programs (28 hours). Delivered at Istituti Tecnici "Vito Sante Longo", Monopoli (BA).
- *2025*
  **Training Lecturer** for the course "Fundamentals of Databases" (60 hours). Delivered at ITS Apulia Digital.
- *2023*
  **Training Lecturer** for the course "Laboratory of Computational Neuroscience with Brain-Computer Interfaces". Course delivered by DEI, Politecnico di Bari, within PNRR project. Delivered at:
  1. I.T.E.T. "Blaise Pascal", Foggia (4 hours)
  2. I.I.S.S. "G. Ferraris", Molfetta (3 hours)
- *2023*
  **Occasional Consultant** (19/10/2023) - Lecturer for training course "Data driven, Artificial Intelligence and new technologies in the digital transformation of Public Administration". Assinter Italia.

- *2023*
  **Occasional Consultant** (11/05/2023) - Awarded contract for supplementary, preparatory and remedial teaching activities for a.y. 2022-2023 for the course "Deep Learning" (40 hours). Politecnico di Bari.

- *2021*
  **Occasional Consultant** (12/12/2021) - Awarded contract for supplementary, preparatory and remedial teaching activities for a.y. 2021-2022 for the course "Informatics for Engineering" (40 hours). Politecnico di Bari.

- *2022-2023*
  **PhD Student Intern** - PhD student internship under agreement between Politecnico di Bari and the Laboratory of "Methodology and Management of Clinical Experimental Data", IRCCS "Saverio De Bellis", Castellana Grotte (BA). Supervisor: Prof. Rodolfo Sardone.

# 4 LIST OF SCIENTIFIC PUBLICATIONS

## 4.1 Journal Articles


1. Paolo Sorino, Alessandro De Bellis, Daniele Malitesta, Domenico Lofù, Caterina Bonfiglio, Rossella Donghia, Gianluigi Giannelli, Antonio Ferrara, Fedelucio Narducci, Tommaso Di Noia, and Eugenio Di Sciascio. "AMIUgraph: analysis and modeling of interactions for utility-driven benchmarking of graph-based models in healthcare". In: BMC Medical Informatics and Decision Making (2026). URL: https://doi.org/10.1186/s12911-026-03717-5.

2. Paolo Sorino, Angela Lombardi, Domenico Lofù, Tommaso Colafiglio, Antonio Ferrara, Fedelucio Narducci, Eugenio Di Sciascio, and Tommaso Di Noia. "Detecting label noise in longitudinal Alzheimer's data with explainable artificial intelligence". In: Brain Informatics 12.1 (2025), p. 15. URL: https://doi.org/10.1186/s40708-025-00261-2.

3. Domenico Lofù, Paolo Sorino, Tommaso Colafiglio, Caterina Bonfiglio, Rossella Donghia, Gianluigi Giannelli, Angela Lombardi, Tommaso Di Noia, Eugenio Di Sciascio, and Fedelucio Narducci. "MORIX: Machine learning-aided framework for lethality detection and MORtality inference with eXplainable artificial intelligence in MAFLD subjects". In: Computer Methods and Programs in Biomedicine Update (2025), p. 100176.

4. Tommaso Colafiglio, Carmelo Ardito, Paolo Sorino, Domenico Lofù, Fabrizio Festa, Tommaso Di Noia, and Eugenio Di Sciascio. "NeuralPMG: A Neural Polyphonic Music Generation System Based on Machine Learning Algorithms". In: Cognitive Computation 16 (2024), pp. 2779–2802.

5. Tommaso Colafiglio, Angela Lombardi, Paolo Sorino, Elvira Brattico, Domenico Lofù, Danilo Danese, Eugenio Di Sciascio, Tommaso Di Noia, and Fedelucio Narducci. "NeuroSense: A Novel EEG Dataset Utilizing Low-Cost, Sparse Electrode Devices for Emotion Exploration". In: IEEE Access 12 (2024), pp. 159296–159315.

6. Simona Aresta, Ilaria Bortone, Francesco Bottiglione, Tommaso Di Noia, Eugenio Di Sciascio, Domenico Lofù, Mariapia Musci, Fedelucio Narducci, Andrea Pazienza, Rodolfo Sardone, et al. "Combining biomechanical features and machine learning approaches to identify fencers' levels for training support". In: Applied Sciences 12.23 (2022), p. 12350.

7. Francesco Maria Calabrese, Vittoria Disciglio, Isabella Franco, Paolo Sorino, Caterina Bonfiglio, Antonella Bianco, Angelo Campanella, Tamara Lippolis, Pasqua Letizia Pesole, Maurizio Polignano, et al. "A low glycemic index Mediterranean diet combined with aerobic physical activity rearranges the gut microbiota signature in NAFLD patients". In: Nutrients 14.9 (2022), p. 1773.

8. Ritanna Curci, Antonella Bianco, Isabella Franco, Angelo Campanella, Antonella Mirizzi, Caterina Bonfiglio, Paolo Sorino, Fabio Fucilli, Giuseppe Di Giovanni, Nicola Giampaolo, et al. “The Effect of Low Glycemic Index Mediterranean Diet and Combined Exercise Program on Metabolic-Associated Fatty Liver Disease: A Joint Modeling Approach”. In: Journal of Clinical Medicine 11.15 (2022), p. 4339.

9. Angelo Campanella, Paolo Sorino, Caterina Bonfiglio, Antonella Mirizzi, Isabella Franco, Antonella Bianco, Giovanni Misciagna, Maria Gabriella Caruso, Anna Maria Cisternino, Maria Notarnicola, et al. “Effects of weight change on all causes, digestive system and other causes mortality in southern Italy: A competing risk approach”. In: International Journal of Obesity 46.1 (2022), pp. 113–120.

10. Rossella Tatoli, Luisa Lampignano, Ilaria Bortone, Rossella Donghia, Fabio Castellana, Roberta Zupo, Sarah Tirelli, Sara De Nucci, Annamaria Sila, Annalidia Natuzzi, et al. “Dietary patterns associated with diabetes in an older population from Southern Italy using an unsupervised learning approach”. In: Sensors 22.6 (2022), p. 2193.

11. Paola Zinno, Francesco Maria Calabrese, Emily Schifano, Paolo Sorino, Raffaella Di Cagno, Marco Gobbetti, Eugenio Parente, Maria De Angelis, and Chiara Devirgiliis. “FDF-DB: A Database of Traditional Fermented Dairy Foods and Their Associated Microbiota”. In: Nutrients 14.21 (2022), p. 4581.

12. Antonella Bianco, Isabella Franco, Alberto Rubén Osella, Gianluigi Giannelli, Giuseppe Riezzo, Caterina Bonfiglio, Laura Prospero, Paolo Sorino, and Francesco Russo. “Physical Activity Reduction and the Worsening of Gastrointestinal Health Status during the Second COVID-19 Home Confinement in Southern Italy”. In: International Journal of Environmental Research and Public Health 18.18 (2021), p. 9554.

13. Angelo Campanella, Giovanni Misciagna, Antonella Mirizzi, Maria Gabriella Caruso, Caterina Bonfiglio, Laura R Aballay, Liciana Vas de Arruda Silveira, Antonella Bianco, Isabella Franco, Paolo Sorino, et al. “The effect of the Mediterranean Diet on lifespan: a treatment-effect survival analysis of a population-based prospective cohort study in Southern Italy”. In: International Journal of Epidemiology 50.1 (2021), pp. 245–255.

14. Isabella Franco, Antonella Bianco, Caterina Bonfiglio, Paolo Sorino, Antonella Mirizzi, Angelo Campanella, Claudia Buongiorno, Rosalba Liuzzi, and Alberto R Osella. “Decreased levels of physical activity: results from a cross-sectional study in southern Italy during the COVID-19 lockdown.” In: The Journal of Sports Medicine and Physical Fitness 61.2 (2021), pp. 294–300.

15. Isabella Franco, Antonella Bianco, Antonella Mirizzi, Angelo Campanella, Caterina Bonfiglio, Paolo Sorino, Maria Notarnicola, Valeria Tutino, Raffaele Cozzolongo, Vito Giannuzzi, et al. “Physical activity and low glycemic index Mediterranean diet: Main and modification effects on NAFLD score. Results from a randomized clinical trial”. In: Nutrients 13.1 (2021), p. 66.

16. Antonella Mirizzi, Laura R Aballay, Giovanni Misciagna, Maria G Caruso, Caterina Bonfiglio, Paolo Sorino, Antonella Bianco, Angelo Campanella, Isabella Franco, Ritanna Curci, et al. "Modified WCRF/AICR score and all-cause, digestive system, cardiovascular, cancer and other-cause-related mortality: a competing risk analysis of two cohort studies conducted in Southern Italy". In: Nutrients 13.11 (2021), p. 4002.

17. Paolo Sorino, Angelo Campanella, Caterina Bonfiglio, Antonella Mirizzi, Isabella Franco, Antonella Bianco, Maria Gabriella Caruso, Giovanni Misciagna, Laura R Aballay, Claudia Buongiorno, et al. "Development and validation of a neural network for NAFLD diagnosis". In: Scientific Reports 11.1 (2021), p. 20240.

18. Caterina Bonfiglio, Carla M Leone, Liciana V.A. Silveira, Rocco Guerra, Giovanni Misciagna, Maria G Caruso, Irene Bruno, Claudia Buongiorno, Angelo Campanella, Vito M.B. Guerra, et al. "Remnant cholesterol as a risk factor for cardiovascular, cancer or other causes mortality: a competing risks analysis". In: Nutrition, Metabolism and Cardiovascular Diseases 30.11 (2020), pp. 2093–2102.

19. Angelo Campanella, Palma A. Iacovazzi, Giovanni Misciagna, Caterina Bonfiglio, Antonella Mirizzi, Isabella Franco, Antonella Bianco, Paolo Sorino, Maria G. Caruso, Anna M. Cisternino, et al. "The effect of three Mediterranean diets on remnant cholesterol and non-alcoholic fatty liver disease: a secondary analysis". In: Nutrients 12.6 (2020), p. 1674.

20. Paolo Sorino, Maria Gabriella Caruso, Giovanni Misciagna, Caterina Bonfiglio, Angelo Campanella, Antonella Mirizzi, Isabella Franco, Antonella Bianco, Claudia Buongiorno, Rosalba Liuzzi, et al. "Selecting the best machine learning algorithm to support the diagnosis of Non-Alcoholic Fatty Liver Disease: A meta learner study". In: PLoS One 15.10 (2020), e0240867.

21. Antonella Mirizzi, Isabella Franco, Carla Maria Leone, Caterina Bonfiglio, Raffaele Cozzolongo, Maria Notarnicola, Vito Giannuzzi, Valeria Tutino, Valentina De Nunzio, Irene Bruno, et al. "Effects of some food components on non-alcoholic fatty liver disease severity: results from a cross-sectional study". In: Nutrients 11.11 (2019), p. 2744.

## 4.2 Conference and Workshop Publications

1. Luigi Pio Prencipe, Tommaso Colafiglio, Paolo Sorino, Nadia Giuffrida, Leonardo Caggiani, Fedelucio Narducci, Tommaso Di Noia, and Michele Ottomanelli. "Passenger experience on an automated shuttle service: a survey study on a real test ride in Italy". In: Transportation Research Procedia 95 (2026), pp. 808–815.

2. Domenico Lofù, Paolo Sorino, Tommaso Colafiglio, Angela Lombardi, Tommaso Di Noia, and Fedelucio Narducci. "WeBIUM 2025: 2nd Workshop on Wearable Devices and Brain-Computer Interfaces for User Modelling". In: Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization. 2025, pp. 467–470.

3. Tommaso Colafiglio, Domenico Lofù, Paolo Sorino, Angela Lombardi, Fedelucio Narducci, and Tommaso Di Noia. "Neural Musical Instruments through Brain-Computer Interface and Biofeedback". In: Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization. 2025, pp. 489–494.

4. Paolo Sorino, Domenico Lofù, Alessandro Iannone, Fedelucio Narducci, Eugenio Di Sciascio, and Tommaso Di Noia. "Enhancing EEG-Based Limbs Movement Classification through Advanced Machine Learning Techniques for Motor Imagery". In: 2025 IEEE 5th International Conference on Human-Machine Systems (ICHMS). 2025, pp. 332–337.

5. Domenico Lofù, Gianluca Colonna, Paolo Sorino, Fabio Castellana, Angela Lombardi, Azzurra Ragone, and Rodolfo Sardone. "Assessing the Short-Term Impact of Air Pollution and Socioeconomic Factors on the Overall Mortality of Taranto: an eXplainable Machine Learning Approach". In: 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC). 2025, pp. 4906–4911.

6. Paolo Sorino, Domenico Lofù, Fedelucio Narducci, Tommaso Di Noia, Eugenio Di Sciascio, and Ignazio Lofù. "Explainable Machine Learning for Clinical Prediction of High-Flow Nasal Cannula Therapy in Pediatric Bronchiolitis". In: 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC). 2025, pp. 4912–4917.

7. Carmelo Ardito, Tommaso Colafiglio, Tommaso Di Noia, Angela Lombardi, Domenico Lofù, Fedelucio Narducci, and Paolo Sorino. "Towards a Neurofeedback Tool For Emotion Recognition Using Brain Computer Interface". In: International Conference on Web Engineering. Springer. 2024, pp. 89–98.

8. Tommaso Colafiglio, Tommaso Di Noia, Domenico Lofù, Angela Lombardi, Fedelucio Narducci, and Paolo Sorino. "Wearable Devices and Brain-Computer Interfaces for User Modelling (WeBIUM)". In: Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization. 2024, pp. 597–600.

9. Tommaso Colafiglio, Domenico Lofù, Paolo Sorino, Angela Lombardi, Fedelucio Narducci, Fabrizio Festa, and Tommaso Di Noia. "EmoSynth Real Time Emotion-Driven Sound Texture Synthesis via Brain-Computer Interface". In: Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization. 2024, pp. 616–621.

10. Danilo Danese, Tommaso Di Noia, Angela Lombardi, Domenico Lofù, Fatemeh Nazary, Rodolfo Sardone, and Paolo Sorino. "Integrating eXplainable AI in Healthcare: A Web Application Framework for Advancing the One Health Paradigm". In: International Conference on Web Engineering. Springer. 2024, pp. 77–88.

11. Domenico Lofù, Paolo Sorino, Tommaso Di Noia, and Eugenio Di Sciascio. "Towards a Federated Intrusion Detection System based on Neuromorphic Computing". In: 2024 9th International Conference on Smart and Sustainable Technologies (SpliTech). 2024, pp. 1–5.

12. Paolo Sorino, Giovanni Maria Biancofiore, Domenico Lofù, Tommaso Colafiglio, Angela Lombardi, Fedelucio Narducci, and Tommaso Di Noia. "ARIEL: Brain-Computer Interfaces meet Large Language Models for Emotional Support Conversation". In: Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization. 2024, pp. 601–609.

13. Paolo Sorino, Gianluca Colonna, Domenico Lofù, Tommaso Colafiglio, Angela Lombardi, Fedelucio Narducci, and Tommaso Di Noia. "An Explainable Machine Learning Approach for Heartbeat Classification Through Signal-Based Features". In: 2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC). IEEE. 2024, pp. 1–6.

14. Tommaso Colafiglio, Domenico Lofù, Paolo Sorino, Fabrizio Festa, Tommaso Di Noia, and Eugenio Di Sciascio. "Exploring the Mental State Intersection by Brain-Computer Interfaces, Cellular Automata and Biofeedback". In: IEEE EUROCON 2023-20th International Conference on Smart Technologies. IEEE. 2023, pp. 461–466.

15. Tommaso Colafiglio, Paolo Sorino, Domenico Lofu, Angela Lombardi, Fedelucio Narducci, and Tommaso Di Noia. "Combining Mental States Recognition and Machine Learning for Neurorehabilitation". In: 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC). IEEE. 2023, pp. 3848–3853.

16. Angela Lombardi, L De Bonis, Giuseppe Fasano, Alessia Sportelli, Tommaso Colafiglio, Domenico Lofù, Paolo Sorino, Fedelucio Narducci, Eugenio Di Sciascio, and Tommaso Di Noia. "Time-to-event interpretable machine learning for multiple sclerosis worsening prediction: results from iDPP@ CLEF 2023". In: CLEF. 2023.

17. Paolo Sorino, Vincenzo Paparella, Domenico Lofu, Tommaso Colafiglio, Eugenio Di Sciascio, Fedelucio Narducci, Rodolfo Sardone, and Tommaso Di Noia. "A Pareto-Optimality-based approach for selecting the best Machine Learning models in mild cognitive impairment prediction". In: 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC). IEEE. 2023, pp. 3822–3827.

18. Carmelo Ardito, Ilaria Bortone, Tommaso Colafiglio, Tommaso Di Noia, Eugenio Di Sciascio, Domenico Lofù, Fedelucio Narducci, Rodolfo Sardone, and Paolo Sorino. "Brain computer interface: Deep learning approach to predict human emotion recognition". In: 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC). IEEE. 2022, pp. 2689–2694.

19. Fabio Castellana, Simona Aresta, Paolo Sorino, Ilaria Bortone, Domenico Lofù, Fedelucio Narducci, Tommaso Di Noia, Eugenio Di Sciascio, and Rodolfo Sardone. "An artificial neural network model to assess nutritional factors associated with frailty in the aging population from southern italy". In: 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC). IEEE. 2022, pp. 3228–3233.

20. Tommaso Colafiglio, Paolo Sorino, Angela Lombardi, Domenico Lofù, and Tommaso Di. "Predicting Human Emotions using EEG-based Brain computer Interface and Interpretable Machine Learning". In: (2022).

21. Domenico Lofù, Pietro Di Gennaro, Paolo Sorino, Tommaso Di Noia, and Eugenio Di Sciascio. "Cpu-side comparison for key agreement between tree parity machines and standard cryptographic primitives". In: 2022 12th International Conference on Dependable Systems, Services and Technologies (DESSERT). IEEE. 2022, pp. 1–6.

22. Paolo Sorino. "Blockchain and AI to Build an Alzheimer's Risk Calculator". In: International Conference on Web Engineering. Springer. 2022, pp. 432–436.

# 5 EXTRA-SCIENTIFIC ACTIVITIES

## 5.1 Institutional Academic Service

- *2022–2024* – Elected representative of PhD candidates and research fellows in the Department Board (Consiglio di Dipartimento) and in the Department Executive Committee (Giunta di Dipartimento), Department of Electrical and Information Engineering (DEI), Politecnico di Bari.

# 6 DISSERTATION CO-SUPERVISION

Since 2019, he has co-supervised more than 20 Bachelor's and Master's theses in Computer Engineering and related programmes, mainly focusing on:

- machine learning and deep learning for healthcare and biomedical data (e.g., explainable AI for Alzheimer's disease, ECG and EEG analysis, risk prediction systems);
- brain-computer interfaces and human–computer interaction;
- data-driven decision support systems and recommender systems;
- design and development of web and e-commerce platforms and database applications.

Selected theses:

- "Explainability in Alzheimer's Disease Diagnosis: A Deep Learning Approach with Neuroimaging Data" (MSc thesis) – Supervisor: Prof. Tommaso Di Noia; Co-supervisors: Eng. Paolo Sorino, Eng. Domenico Lofù.

- "Enhancing EEG-based limbs movement classification through advanced machine learning techniques for motor imagery" (MSc thesis) – Supervisor: Prof. Tommaso Di Noia; Co-supervisor: Eng. Paolo Sorino.
- "Explainability in Alzheimer's Disease Diagnosis: A Deep Learning Approach with Neuroimaging Data" (MSc thesis) – Supervisor: Prof. Tommaso Di Noia; Co-supervisor: Eng. Paolo Sorino, Eng. Domenico Lofù.
- "Sviluppo di una web application per il monitoraggio e l'analisi delle prestazioni degli atleti" (BSc thesis) – Supervisor: Prof. Antonio Ferrara; Co-supervisors: Prof. Carmelo Antonio Ardito, Eng. Paolo Sorino.
- "Progettazione di una rete neurale convoluzionale per il riconoscimento di segnali EEG" (BSc thesis) – Supervisor: Prof. Giovanni Pascoschi; Co-supervisor: Eng. Paolo Sorino.

# 7 SKILLS

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## 7.1 Technical Skills

**Programming Languages:**

- **Expert**: Python, Java, C/C++, SQL
- **Proficient**: HTML, CSS, JavaScript, PHP, Matlab

**AI and ML Frameworks:**

- TensorFlow, Keras, PyTorch, Scikit-learn
- Pandas, NumPy, SciPy, Matplotlib, Seaborn
- XAI tools: SHAP, LIME, Captum

**Development Environments and Tools:**

- IDEs: Visual Studio, PyCharm, Eclipse, IntelliJ IDEA
- Version Control: Git, GitHub
- LaTeX for scientific writing
- Jupyter Notebooks, Google Colab

**Database Systems:**

- SQL databases: MySQL, PostgreSQL, Oracle, MSSQL
- NoSQL databases: MongoDB

**Web Development:**

- Frameworks: WordPress, Django, Flask
- Technologies: HTML5, CSS3, JavaScript, PHP

**Operating Systems:**

- Linux (Ubuntu, Debian, CentOS)
- Windows
- macOS

**Software and Tools:**

- Office: Microsoft Office, LaTeX
- Image editing: Photoshop, Lightroom
- Data analysis: STATA, R

## 7.2 Communication Skills

Excellent communication and interpersonal skills acquired in professional and academic environments. Active in sports (beach volleyball, volleyball) and parish youth activities, which has offered opportunities to meet diverse people and build constructive relationships. Exposure to work and university contexts and their related challenges has contributed to shaping a versatile personality, ready to face a wide range of situations.

Strong presentation skills demonstrated through numerous invited talks and conference presentations at international venues.

## 7.3 Organizational and Managerial Skills

Strong teamwork coordination skills acquired during university studies and research activities. Demonstrated leadership abilities in various contexts, including:

- Work-Package Leader responsibilities in multiple national research projects (OMIBREED, CALLIOPE, REACH-XY)
- Workshop organization and program committee chair roles at international conferences
- Supervision and coordination of 25+ thesis students
- Sports team management (beach volleyball coaching and organization)

Experience in project management, deadline definition, and coordination of inter-institutional activities involving multiple partners.

## 7.4 Leadership Experience

Representative of PhD students and Research Fellows of the Department of Electrical and Information Engineering, Politecnico di Bari, for the term 2022-2024 in both the Department Council and the Department Board (as reported by collaborators).

Active community organizer for Beach & Chill sports group in Monopoli area, managing tournaments and training activities.

## 7.5 Language Skills

- **Italian**: Native language
- **English**: C1 level (Listening, Reading, Interaction, Oral Production, Written Production)
- **French**: A2 level (Listening, Reading, Interaction, Oral Production, Written Production)

### 7.6 Other Skills

Photography, Sports (beach volleyball, volleyball), Music, Cinema, Art, Pokemon card collecting, Travel and exploring historic sites.

### 7.7 Driving License

A, B

## Statement

**Authorization for the processing of personal data.** I hereby declare that the information provided in this CV is true, complete and accurate to the best of my knowledge. I authorize the processing of my personal data for recruitment and evaluation purposes, in accordance with applicable data protection regulations (EU Regulation 2016/679 - GDPR and Italian Legislative Decree 196/2003 as amended).

The undersigned is aware that, in case of false declarations, criminal sanctions referred to in art. 76 of Presidential Decree 445/2000 and subsequent amendments apply, as well as the forfeiture of any benefits obtained on the basis of untrue declarations.

Monopoli, July 29, 2026

Paolo Sorino

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